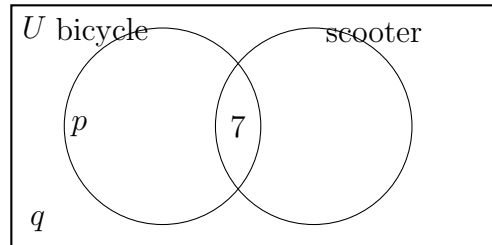


4.12 Quiz: Probability

- In a survey of 40 students, 22 own a bicycle and 15 own a scooter. Seven students own both, as shown in the following Venn diagram. The values p and q represent numbers of students.



- Find the value of p . [1]
- Find the value of q . [1]
- A student is chosen at random. Find the probability that the student owns a scooter but not a bicycle. [2]

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- Events A and B are mutually exclusive with $P(A) = 0.55$ and $P(B) = 0.15$.

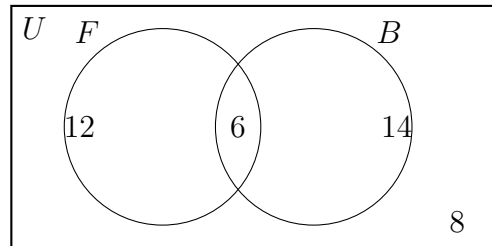
- Write down $P(A \cap B)$. [1]
- Find $P(A \cup B)$. [2]

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3. In a group of 40 students, some play football (F) and some play basketball (B). Eight students play neither sport. This information is shown in the following Venn diagram.



- (a) Write down the number of students in the group who play football. [2]
- (b) One student from the group is chosen at random. Find the probability that
 - (i) the student does not play football;
 - (ii) the student plays football or basketball, but not both. [4]